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DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

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"A" Grade (CGPA 3. lei

Department of Artificial Intelligence and Machine Learning

B.Tech. Sem: V Subject: Fullstack Development with NextJs (DJS23AMD302)

EXPERIMENT NO. 1

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AIM

To scaffold a modern Next.js App Router workspace utilizing TypeScript, implementing file-system routing across nested directories, dynamic route segments ([i d)), route ou s, and isolated React Server Corn onent RSC vs Client Com onent boundaries.

THEORY

App Router Architecture: Next.js uses an app/ directory model that leverages React Server Components (RSC) by default. This enables automatic server-side rendering, partial page hydration, and zero client-side JavaScript overhead for non-interactive UI subtrees. File-System Routing Rules: Directories define route paths, while specific reserved files govern layout inheritance and page rendering:

page . tsx: Defines the unique UI rendered for a specific URL route segment.

layout . tsx: Defines persistent, structural UI wrapper frames shared across child routes without re-mounting state.

[Idl /page . tsx: Dynamic route segments that capture URL params.

(group) / : Route groups that organize files into logical sub-trees without altering the URL path.

Server vs. Client Components: Components render on the server by default. Interactive components requiring browser APIs, React hooks (useState, useEf feet), or DOM event handlers must ex licitl declare the 'use client ' directive at the t ofthe file.

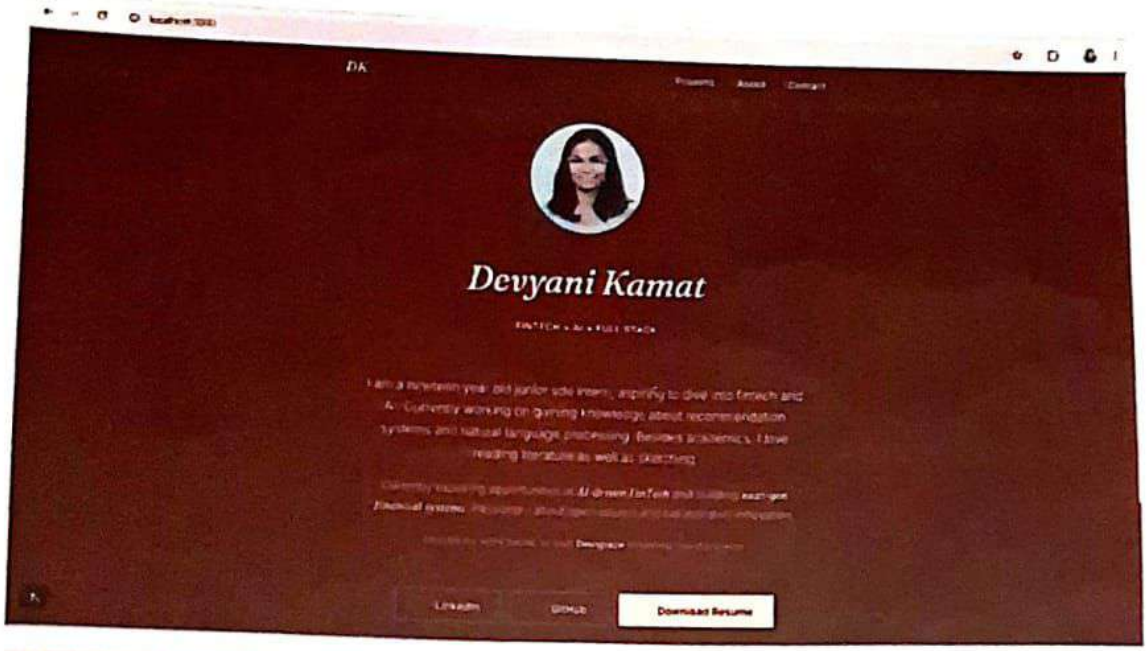
OBSERVATION

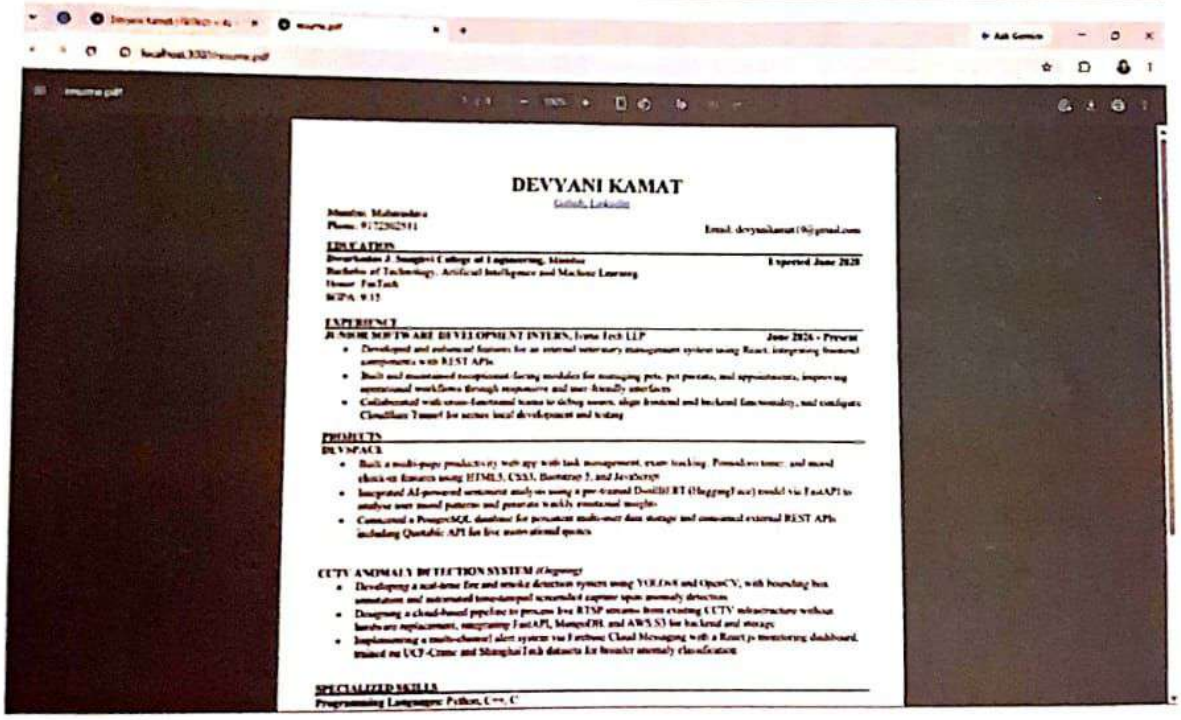
A nextjs project was successfully created using the app router & typescript file based routing nested routes, dynamic route segments, route groups, & the destination between servers & client components were implemented & tested successfully.

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OUTPUT





CONCLUSION

The experiment helped understand the app router architecture & routing system in Next.js enabling the development of modern & scalable web application.